

Assessing the Conservation Status of Invasive *Acomys cilicicus*: Conservation Taxonomic Unit or Distinct Species?

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Abstract

Acomys cilicicus is a rodent species endemic to Turkey, with a highly restricted distribution. However, there have been doubts regarding its taxonomic validity as a distinct species. Previous studies investigating the phylogeny of *A. cilicicus* have been unable to provide conclusive data to address this validity issue, primarily due to their reliance on two specimens from a single locality. In this study, we expanded the sample size to include 16 specimens from 5 different localities and conducted sequencing of a mitochondrial DNA gene region (cyt-b). Initial analysis of genetic distances revealed a close relationship between *cilicicus*, *minous*, and *nesiotes*. To further explore the phylogenetic relationships, we incorporated additional mtDNA sequences from GenBank into our analysis, resulting in the construction of a phylogenetic tree. Our findings demonstrate that *cilicicus* forms a distinct clade, supporting its status as a separate species. Moreover, the *cilicicus* clade was identified as the sister clade of *A. minous*. By estimating the divergence time, we determined that *cilicicus* diverged from its closest relative, *minous*, approximately 170 thousand years ago. Additionally, based on our sequence data, we propose a hypothesis suggesting that *cilicicus* might have been introduced to Anatolia from Crete during the Lower Palaeolithic era. We further discussed the inferred migration pathways of the Mediterranean region based on clade formation and migration analysis.

1. Introduction

When a species invades a new area, it does not necessarily mean that the invasive species will live there. If the founder population is not fit enough to survive, extinction is inevitable (Abdelkrim et al., 2005). Even after a successful invasion, population genetic parameters may signal extinction. The study on Guadeloupe Island rats was a good example (Abdelkrim et al. 2005). Genetic diversity must be high enough to support fitness for survival (Lee, 2002). Genetic diversity is one of the main parameters affecting the fate of a population on the path to extinction, besides competition with other species and the level of adaptation to a new habitat (Lee, 2002). In addition to genetic challenges in the modern world, the invasive species face additional hurdles that need to be overcome, including the integrated efforts of humans under the One Health approach or through agricultural practices, combating them with rodenticides, and addressing the impacts of habitat destruction and other environmental impacts (Castano *et al.* 2023). In natural way there might be two reasons for the loss of genetic diversity, which is a sign of a bottleneck: either the founder population had low genetic diversity or genetic drift occurred in the invasion area. Therefore, determining the genetic diversity of an introduced species is important. The introduction of a population may also lead to speciation. *Acomys* is a good candidate for studying this hypothesis: humans traveling in the Mediterranean and introducing this animal to places they visited may have resulted in the evolution of new species, since *Acomys* has 26 molecular taxonomic units and all of them can be distinct species (Aghova *et al.* 2019).

The genus *Acomys* is distributed throughout the Arabian Peninsula, Africa, Southern Turkey, Cyprus, and Crete (Bates, 1994). The genus is represented with 19 species in these localities (Wilson and Reeder, 2005). Aghova *et al.* (2019) grouped the genus into five main groups (*subspinosus*, *spinosissimus*,

russatus, *wilsoni*, and *cahirinus*). They emphasized that the Mediterranean species were placed in the *cahirinus* group and that the status of the species was doubtful, and detailed analyses were suggested. The origin of *A. minous* in Cyprus has been previously mentioned as *A. nesiotus* (Barome et al. 2001). The discussion about the species status of *Acomys* on islands and Turkey was continued by Giagia-Athanasopoulou et al. (2011). The low genetic diversity between the three species (*nesiotus*, *minous* and *cilicicus*) was prominent in the discussion, and they claimed that the karyotype was not a taxonomic character to separate species. It has been shown with geometric morphometric and genetic analyzes that the distributed species of *Acomys* in the Mediterranean were introduced at different times, detailed analyzes were incomplete (Renaud et al. 2020). Frynta et al. (2010) noted that *A. cilicicus*, *A. nesiotus*, and *A. minous* were commercially introduced to all distribution areas in the Mediterranean islands and Anatolia. The first *Acomys* record from Turkey was given by Lehmann (1969) from Silifke, and initially as *A. cahirinus*. Later, the same researcher designated the same population as part of *A. nesiotus* (Lehmann, 1969). Spitzenberger (1978) compared the morphological peculiarities of specimens from Silifke with those of *A. nesiotus* from Cyprus and *A. minous* from Crete and classified them as a new species, *Acomys cilicicus*, based on differences in incisors, molar characteristics, hind feet, and body measurements. Sözen et al. (2017) defined the exact distribution of these species, which was only an area of 120 square kilometers along the Mediterranean coast of Turkey. They also observed that the only rodent with *A. cilicicus* was *Apodemus mystacinus*.

We aimed to reveal the phylogeographic structure and biogeographic history of *Acomys* species, especially *A. cilicicus*, by working out the phylogeny and estimating the separation time so that it will help to resolve whether *Acomys cilicicus* is a valid species. We also discussed the estimations of the separation times of *cilicicus*, its arrival in Anatolia, and supporting data about the sea voyages of the genus *Homo* on these dates.

2. Materials and Methods

A. cilicicus specimens (n = 16) were collected from Mersin (Table 1, Fig. 1); Trapped with Sherman traps, tissue was taken from the ear and liver. Later, to reveal the phylogeny of species, DNA samples were isolated from tissue using a 5-Prime DNA isolation kit. We sequenced *cyt-b* genes from mtDNA using the primers L14724 5'-GATATGAAAACCATCGTTG-3' and H15915 5'-TCTCCATTTCTGGTTTACAAGAC-3' (Irwin et al. 1991). Raw *cyt-b* gene sequences were corrected and a consensus sequence was created for each individual specimen via Geneious; mismatches or misreadings were manually corrected. We also used previously published sequences belonging to the other species to align the sequences (appendix Table 1). All sequences were aligned via Clustal X using BioEdit. After alignment processes, some parts of the sequences were deleted since most parts were covered gaps and misread manually. We uploaded 16 sequences to GenBank (in appendix, Table 1), which ranged for 950 bp.

Table 1

Sampling locations within the distribution area of *Acomys cilicicus*; their coordinates, altitudes (m), localities (+), and haplotypes revealed from *cyt-b* sequences.

Localities (no in fige 1)	Coordinates	Altitude (m)	N	Haplotype			
				H1	H2	H3	H4
Karahmetli (27, 28, 29)	36°31'38.82"N, 34°7'26.52"E	483–630	3	1	1	1	0
Narlıkuyu (17,19)	36°27'7.89"N, 34°6'59.17"E	81–108	7	5	1	0	1
Hüseyinler (15)	36°30'59.02"N, 34°1'50.75"E	720 m	1	1	0	0	0
Ayaş (24)	36°29'33.21"N, 34°9'44.65"E	117–196	3	3	0	0	0
Kumkuyu (32)	36°32'51.17"N, 34°10'35.28"E	347–387	2	2	0	0	0

A median joining network was drawn using Network software. (<https://www.fluxus-engineering.com/sharenet.htm>). To draw networks, roehl data files were prepared in DNAsp (Librado and Rozas, 2009). Sequences from the genbank were also included in the network to better understand the relationships between *cilicicus* (Suppl. Table 1). Also, we provide a Kimura-2 model distance calculated in MEGA 11 (Tamura et al, 2011).

Phylogeny was reconstructed using maximum likelihood (ML) approaches. The reconstruction substitution model (TIM2 + I) was chosen using jModelTest 2 (Darriba et al. 2012) applying Corrected Akaike Information Criterion (AICc) weights as the selection criterion via online tools Cipres portal (<https://www.phylo.org/>). The ML tree was reconstructed using the web server PHYML (Guindon et al. 2010), which was used as an online tool <http://www.atgc-montpellier.fr/phyml/>. We visualized all trees with FigTree v.1.4 (Rambaut, 2012).

Separation time estimations were estimated by BEAST v2.5, and the best fitting substitution model was selected using jModeltest-2 (Darriba et al. 2012 via online tool phylo.org/cipres). The estimation parameters were optimized in BEAUTI 2.5. The analysis was executed with Optimized relaxed molecular clock and a calibrated Yule prior. Ten replicate MCMC chains were performed each for 100 million generations with the effective sample size (ESS) obtained for each parameter above 200. Posterior distributions of MRCAs and other model parameters were checked with TRACER 1.5 (Rambaut & Drummond, 2009), and posterior samples from all chains were combined in LogCombiner, which is a part of the BEAST package. The Maximum Clade Credibility (MCC) tree was prepared using TreeAnnotator, also a part of the BEAST package, and was drawn in FigTree 1.4 (<http://tree.bio.ed.ac.uk/software/figtree/>) to reveal the species status of our sampling of *Acomys* in the Mediterranean Basin. We estimated the direction and magnitude of gene flow between pairs of populations with different sizes using a coalescent-based maximum likelihood approach for the purpose of estimating long-term gene flow (Beerli and Felsenstein 2001) using Migrate 3.6.6 (Beerli, 2009). Parameters were estimated from data. We set the number of recorded steps in a chain as 1,000. Three

possible scenarios for three populations (*A. minous*-Crete as 1st, *A. cilicicus*-Turkey as 2nd, and *A. cahirinus*-Egypt as 3rd) were tested.

3. Results

All sequences were uploaded to NCBI with the accession numbers OP791975- OP791977 (Suppl Table 1). The network revealed that the relationships between haplotypes and two haplotypes might be endemic haplotypes because they are placed separately (Fig. 2, Table 1, and in Suppl. Table 1, 2). It also revealed that Cyprus spiny mice and Silifke spiny mice grouped with different haplotypes from Crete. *minous*, became an ancestral species for *cilicicus* and *nesiotes*, and the network revealed that *nesiotes* and *cilicicus* derived from different haplotypes of *minous*. Also, the kimura-2 distance determines the distance difference within *cilicicus*- like networks (Suppl. Table 3). K-2P distance analysis revealed a low genetic distance between all three study species of *Acomys*. Also, the distance between the *spinosissimus* and *cahirinus* groups was smaller than expected.

According to the maximum likelihood tree, *A. cilicicus* was displayed as a separated branch from *A. minous*, *A. nesiotes*, and *A. cahirinus* (Fig. 2).

We also calculated the separation time of *A. cilicicus* because no data were present for *A. cilicicus* regarding its separation time or phylogeographical pattern. Two calibration points from Aghova et al (2019) were used during the analysis: *spinosissimus*-*subspinosus* (7.04 my) divergence time and the other one between *cahirinus* (Arabian region, cah10 in Aghova et al. 2019) and Northeastern *cahirinus* (Sahara region, cah9 in Aghova et al, 2019) (1.59 my). Our estimations showed that the divergence time between *cilicicus* and its closest ancestral species, *minous*, was approximately 170 k years (Table 2).

Table 2
Separation times (mya) between *Acomys* species. * indicates calibration points.

Seperation point	Seperation time (%95)
<i>cilicicus/ minous</i>	0.17 (0.031, 0.368)
<i>minous/cahirinus</i>	0.26 (0.042, 0.58)
<i>spinosissimus/subspinosus</i> *	7.04 (6.95, 7.17)
<i>cahirinus</i> (cah9)/ <i>cahirinus</i> (cah10)*	1.57 (1.50, 1.64)

We also worked on modeling a possible migration route by MIGRATE. The program tested 3 possible migration routes (Fig. 4). All three possibilities accepted *cahirinus* as the ancestral species, and that Silifke, Crete, and Cyprus were invaded by *Acomys* at different times. Our first model is indicated by the dashed arrow in Fig. 4, which shows all three localities (Silifke - *A. cilicicus*, Crete - *A. minous*, and Egypt - *A. cahirinus*). The first model accepted that *cahirinus* spread to Crete and Silifke separately, and the speciation of *minous* and *cilicicus* at those localities happened independently. The second model (single

headed arrow in Fig. 4) accepted that *cahirinus* first moved to Crete, where it evolved to *minous*, and later moved from Crete to Silifke. The last model accepted that all species moved in a bidirectional fashion; between Crete and Silifke, Silifke and Egypt, as well as between Egypt and Crete (double headed arrows in Fig. 4). According to our results, the second model had the highest support score among all three model probabilities (Table 3).

Table 3
Log marginal likelihood results and Bayesian factors for three models.

Model	Bezier lml	Harmonic lml	LBF (Bezier)	Choice (Bezier)	Model probability
1	-4904.276518	-4607.564427	-42.54	1	0.000
2	-4704.879579	-4489.106412	0.00	2 (best)	0.545
3	-4558.201211	-4392.242926	-3.22	3	0.123

4. Discussion

Acomys cilicicus was described from Silifke, Mersin by Spitzenberger (1978). Until now, we had fairly limited knowledge about its phylogeny, and even its taxonomic status was unclear (Spitzenberger, 1978; Macholán et al. 1995; Kivanç et al. 1997; Barone et al. 2001; Kryštufek and Vohralík, 2009; and Kivanç et al. 2013 Aghova et al. 2019, Renaud et al. 2020). A recently published study determined its distribution in the same region (Çetintaş et al. 2018). The distribution of this species was very narrow and under considerable anthropogenic pressure, and thus the genetic distance calculations resulted in low genetic diversity both within the species and among subpopulations. We reveal two haplotypes. These two haplotypes (Karahmetli and Narlıkuyu) were separated from the others in all analyses. While the genetic distance calculations within *Acomys cilicicus* showed relatively small variation, the distance between the two studied populations of *Acomys cilicicus* was found to be notably higher than the distances between *Acomys cilicicus* and the other two species, *Acomys minous*. This suggests that the genetic divergence between these two localities of *Acomys cilicicus* is comparatively significant. Based on the genetic distance analyses, it may not be supported as a distinct species; however, the indication of their location as a separate branch in both phylogenetic tree topology and network analysis, combined with the presence of geographical isolation and reproductive isolation, supports considering them as separate species. Additionally, the comparison of genetic distances between *subspinosus*, *spinosissimus*, and *cahirinus*, which were used as an outgroup, revealed a much lower divergence compared to the distances observed between species in other taxa, with a value close to 2%, which is commonly considered as the threshold for speciation. This suggests that the rate of genetic differentiation within this group of

Acomys species is relatively slow. Furthermore, according to the chromosome number analysis, while *Acomys cilicicus* originated from *Acomys minous* (Crete), the chromosome number of *Acomys cilicicus* was found to be the same as that of *Acomys cahirinus* ($2n = 36$) and no population with $2n = 36$ chromosomes was found in Crete. This also suggests the presence of geographical isolation or reproductive isolation. Based on observations made in the field, the species is relatively sedentary with a highly restricted distribution range in Anatolia, which may contribute to the slow rate of genetic differentiation. All these indications support calling them different species.

Our phylogenetic tree showed that the closest ancestral species of *A. cilicicus* was *A. minous*, and the bootstrap values supported these results. The tree also showed that both species originated from *cahirinus*. Aghová et al. (2019) grouped the spiny mouse in five major clades: *subspinosus*, *spinosissimus*, *russatus*, *wilsoni*, and *cahirinus*. They placed the Mediterranean species within the *cahirinus* clade (cah9). They also suggested that *cilicicus* was a commensal population of *cahirinus*. The migration analysis results provided three possible scenarios for the migration route of Mediterranean spiny mice with radiation from *cahirinus*. The analysis resulted in a route in which *cahirinus* first invaded Crete, where it evolved to *minous*. The second wave occurred from Crete to Anatolia through an invasion by *minous*. After this second wave of invasion to Anatolia, the evolution and spread of *cilicicus* occurred. *Acomys* originated in southern Africa and dispersed northward by speciation (Aghová et al. 2019). In the northern part of Africa, *cahirinus* was derived around 9 mya. Our molecular clock estimated that the divergence time between *cahirinus* (cah10) Ethiopia and *cahirinus* (cah9) Egypt was 1. *cahirinus/minous* (Table 2, Add. Figure 4) separation as 1.6 mya, which was earlier than the estimation made by Barome et al. (2001), which was based on a 650-bp long data set. Giagia-Athanasopoulou et al. (2011) stated that colonization might have occurred earlier than 0.4 mya by evaluating chromosome variation results combined with *cyt-b* data. Later, the most recent study on *Acomys* (Aghová et al. 2019) dated the split time of *cahirinus* (cah9)/*cahirinus* (cah10) to approximately 1.7 mya and *minous/cahirinus* 1.59 mya, which aligned with our results and pointed out that *cahirinus* was the ancestral species for Mediterranean spiny mice. *nesiotes* could not be grouped as a separate species, probably due to the insufficient number of specimens and the very recent separation of the species. For this reason, detailed analysis is required to understand the status of *nesiotes*.

The migration analysis results showed that both *cilicicus* and *nesiotes* evolved from *minous*, so that all Mediterranean species were grouped within the *cahirinus* clade in our tree as revealed by other studies (Barome et al. 2001, Aghová et al. 2019). Also, the tree given in Barome et al. (2001) placed *minous* and *nesiotes* within a group that was displayed as a sister group of *minous* and *cilicicus*. A similar chromosome topology variation for Crete spiny mouse was high when compared to those of the other island species. This might be the consequence of rapid chromosomal rearrangement and fast fixation of new rearrangements since the effect of a small population on selection appears faster. In other scenarios, the colonization occurred in multiple waves because *nesiotes-minous* and *cilicicus-minous* were grouped into different branches in every study (Barome et al. 2001; Aghová et al. 2019; Giagia-Athanasopoulou et al. 2011; and our results). As estimated by the migration analysis regarding the possible route for colonization of *nesiotes* and *cilicicus*, they originated and evolved from *minous* right after the occupation

of Crete by humans, and because of introduction followed by speciation. The species limitation analysis also proved that *minous* become sources for the two different species in the Mediterranean region. Multiple species delimitation analysis showed that there were two divergence times from *minous* at two dates. One of them aroused *cilicicus* while the other was *nesiotes*. The source of that was probably the invasion of *Acomys* with human movements from different places from the Mediterranean coasts of Africa, where *cahirinus* was distributed, to Crete. Castaneda *et al.* (2010) reported that the invasion of Santa Catalina Island by *Peromyscus fraterculus* resulted in the better adaptation of the invasive species compared to the native species *P. slevi*. Although the three species of *Acomys* found in Mediterranean islands form separate groups in phylogenetic trees, some individuals of the ancestral species, *minous*, enter other branches according to the Migrate analysis. This suggests that populations introduced by humans to Cyprus and Mersin have different levels of adaptation to their respective origins. Kaya Ozdemirel *et al.* (2022) modeled the potential distribution areas of *cilicicus*; however, during our fieldwork since 2002, no individuals belonging to this species were captured (unpublished personal data). This may indicate a low adaptive capacity associated with the introduction of *cilicicus* by humans. This situation raises another question: is the low adaptive capacity of *cilicicus* due to its low genetic diversity resulting from human transportation, or is it because the species is naturally docile, and therefore intentionally brought aboard ships by humans? Uncertainty remains regarding whether *cilicicus* was consciously introduced by humans due to its docile nature. Renaud *et al.* (2022) conducted an analysis showing that *cilicicus* has the smallest skull size among the species found in Mediterranean islands, and when captured in the field, this species exhibits a calm demeanor (unpublished personal observation between 2015–2018). Compared to the sympatric species *Apodemus mystacinus*, *cilicicus* is significantly less mobile in trap-closure areas. Consequently, alongside its low genetic diversity, the presence of a docile species raises suspicions that it may have been intentionally transported by humans on ships, potentially serving as a valuable food source during long sea voyages.

Gafney (2021) reviewed and stated that seafering in the Mediterranean may have started during the Pleistocene from North Africa (figure). In any case, as Barome *et al.* (2001), Giagia-Athanasopoulou *et al.* (2011), and Aghová *et al.* (2019) emphasized *minous* was introduced by humans to Crete, as a descendant of *cahirinus* from Egypt. Humans unintentionally introduced this species (Barome *et al.* 2001), and the colonization might have happened in either a single wave or multiple waves (Giagia-Athanasopoulou *et al.* 2011). Multiple waves seem more likely, considering the chromosomal variation of The Crete spiny mouse. If multiple waves were indeed the case, another question spawned: “Did humans really carry along the spiny mouse unintentionally?”. The absence of *Mus* and other *Rattus* species in the study area has led to the emergence of this issue. Because *Mus* and *Rattus* species are small mammals that are frequently transported during invasion events, often leading to the eradication of other species in the introduced area (Harris 2008). In fact, genetic analyses of the *Mus* genus have shed light on human movements in the Mediterranean (Rodrigues *et al.* 2018). However, our molecular clock calculations have shown that *cilicicus* was introduced to Anatolia earlier. During fieldwork conducted between 2015 and 2018, no other species were found in the area (Çetintaş *et al.* 2019). The introduction of *Acomys* to Crete indicates a much older history than the arrival of *Mus*, as calculated by Cucchi *et al.* (2005) (Table 2). In

light of this evidence, it appears plausible that *Acomys* was transported by humans during the time periods indicated by molecular clock calculations. However most archaeologists and paleoanthropologists believed that long distance hominin migration must have occurred across land and land bridges (Anton and Swisher, 2001; Bar-Yosef and Belfer-Cohen, 2001). Gafney (2021) reviewed the potential harbors during the Pleistocene. Early hominid in North Africa may have invaded the island in the Mediterranean as our migrate analysis suggested (Fig. 4).

The colonization pattern of the spiny mouse was similar to that of the caprine and house mouse, including a stopping station in the Mediterranean. However, from Crete, the transition to Cyprus and Turkey are showing a similar story in *Mus*. In Crete Barome et al (2001) noted *Acomys minous* b haplotype A turkey, a haplotype is the kind of resources that has emerged in Cyprus and our results also support this.

As a result, the occupation of Crete in the Mediterranean occurred at different times and only certain routes were traveled from the cities established here. For this reason, *cilicicus* probably originated from *minous*'s haplotype b, while *nesiotes* probably originated from haplotype a. We have shown this with migrate analysis. We show *cilicicus* as a separate species in both the phylogenetic tree by settling in separate branches and supporting these branches. Although it emerges as a separate species, partial microsatellite results and population genetics analysis performed with *cytb* sequences revealed genetic similarities within *cilicicus*; therefore, we suggest more detailed genetic analysis (SNP, ddRAD) should be performed to determine the genetic structure of *cilicicus* and other Mediterranean species.

Declarations

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Conflict of Interest

Author declares there isn't any conflict of interest.

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Figures

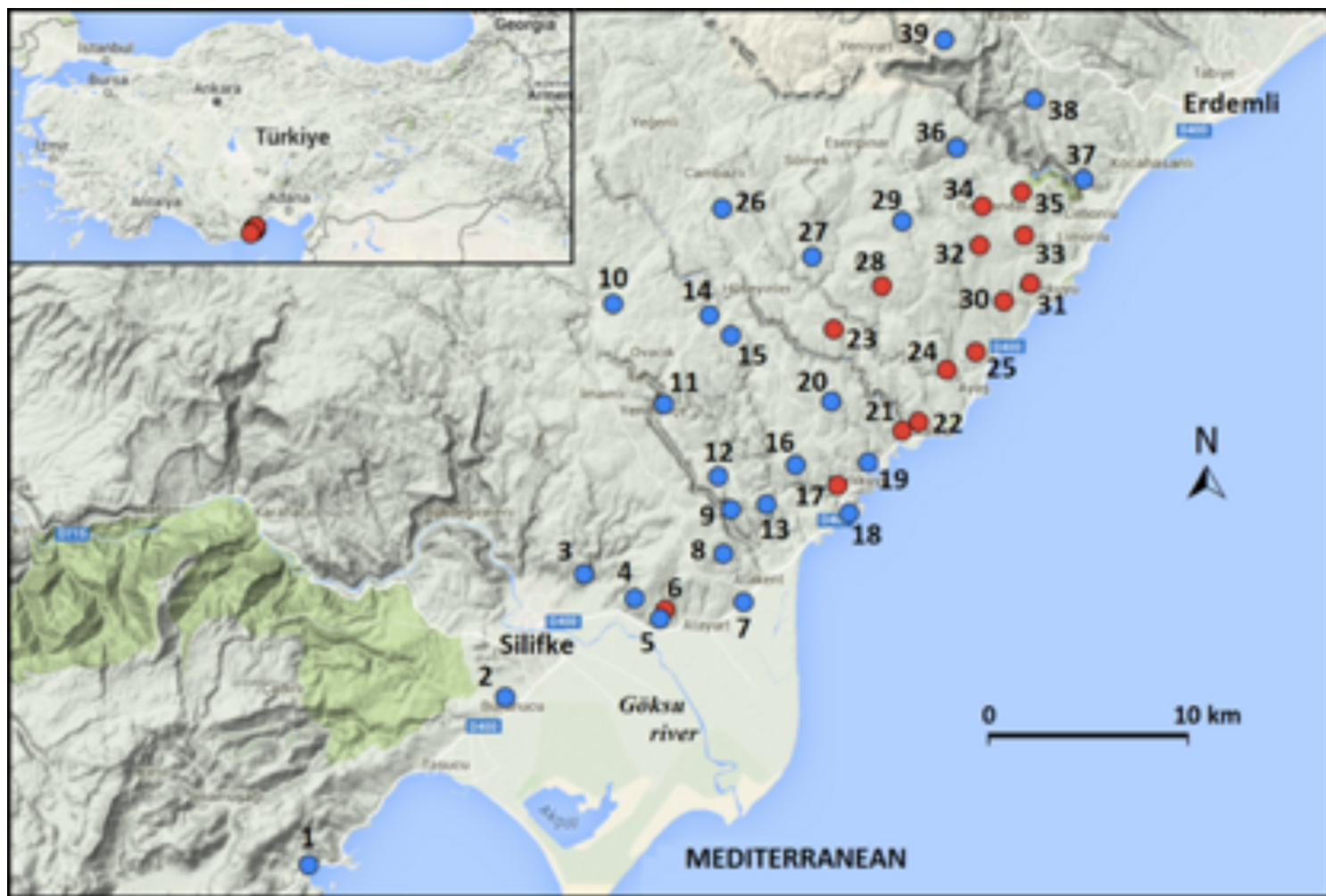


Figure 1

Trapping localities to determine the distribution area of *Acomys cilicicus*. The numbers correspond to those given in Table 1. Blue dots indicate the localities where *Acomys* specimens could not be captured, and red dots indicate the localities where specimens were captured.

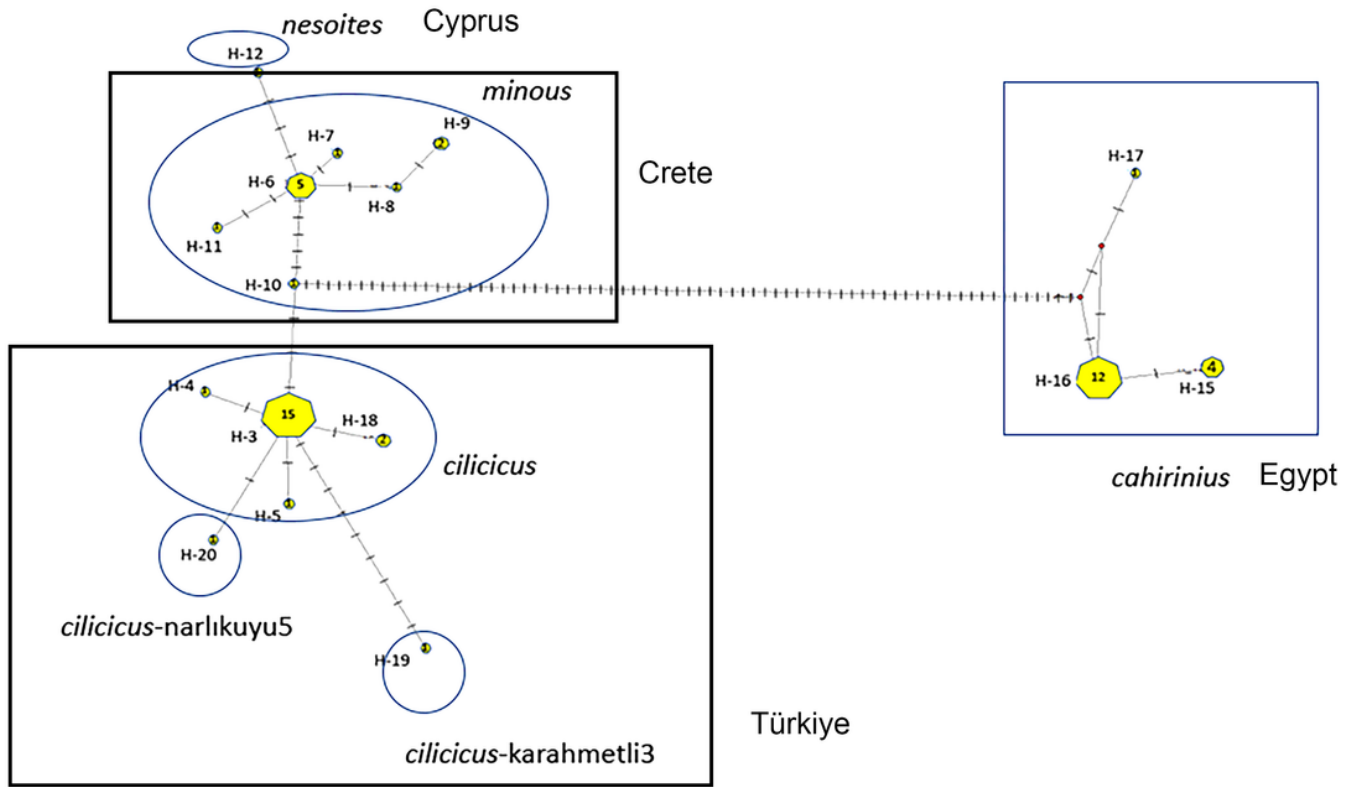


Figure 2

Trapping localities to determine the distribution area of *Acomys cilicicus*. The numbers correspond to those given in Table 1. Blue dots indicate the localities where *Acomys* specimens could not be captured, and red dots indicate the localities where specimens were captured.

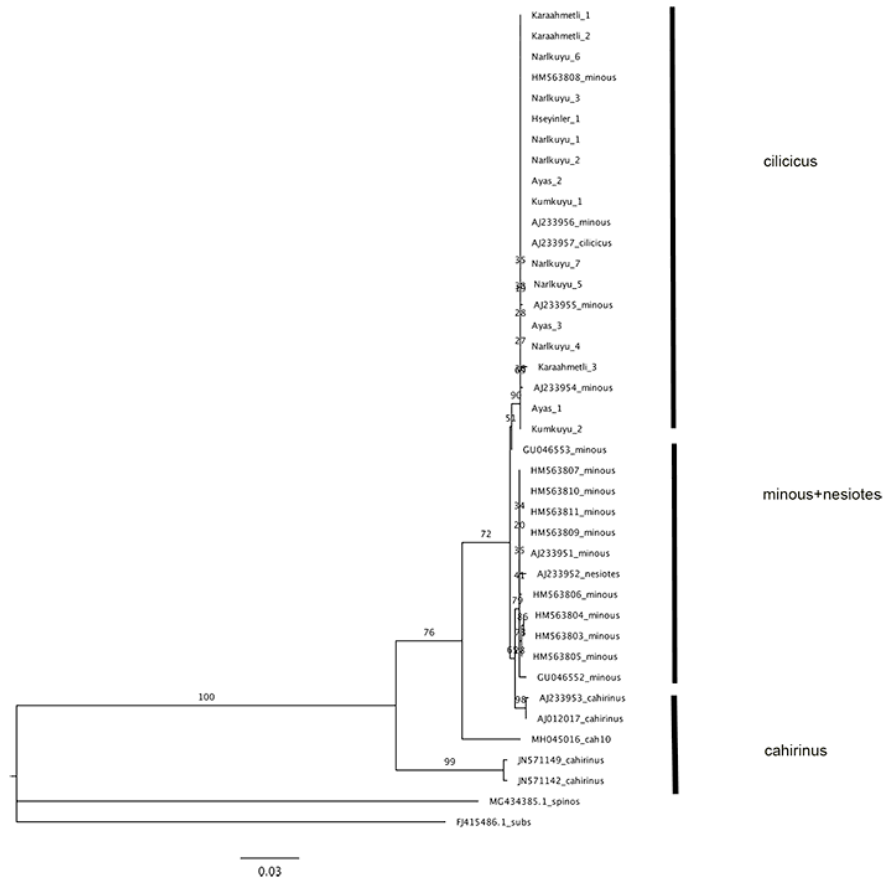


Figure 3

ML tree of *Acomys* species inferred from *cyt-b* gene sequences. Species, including *cilicicus*, are separated from each other with high branch probabilities.

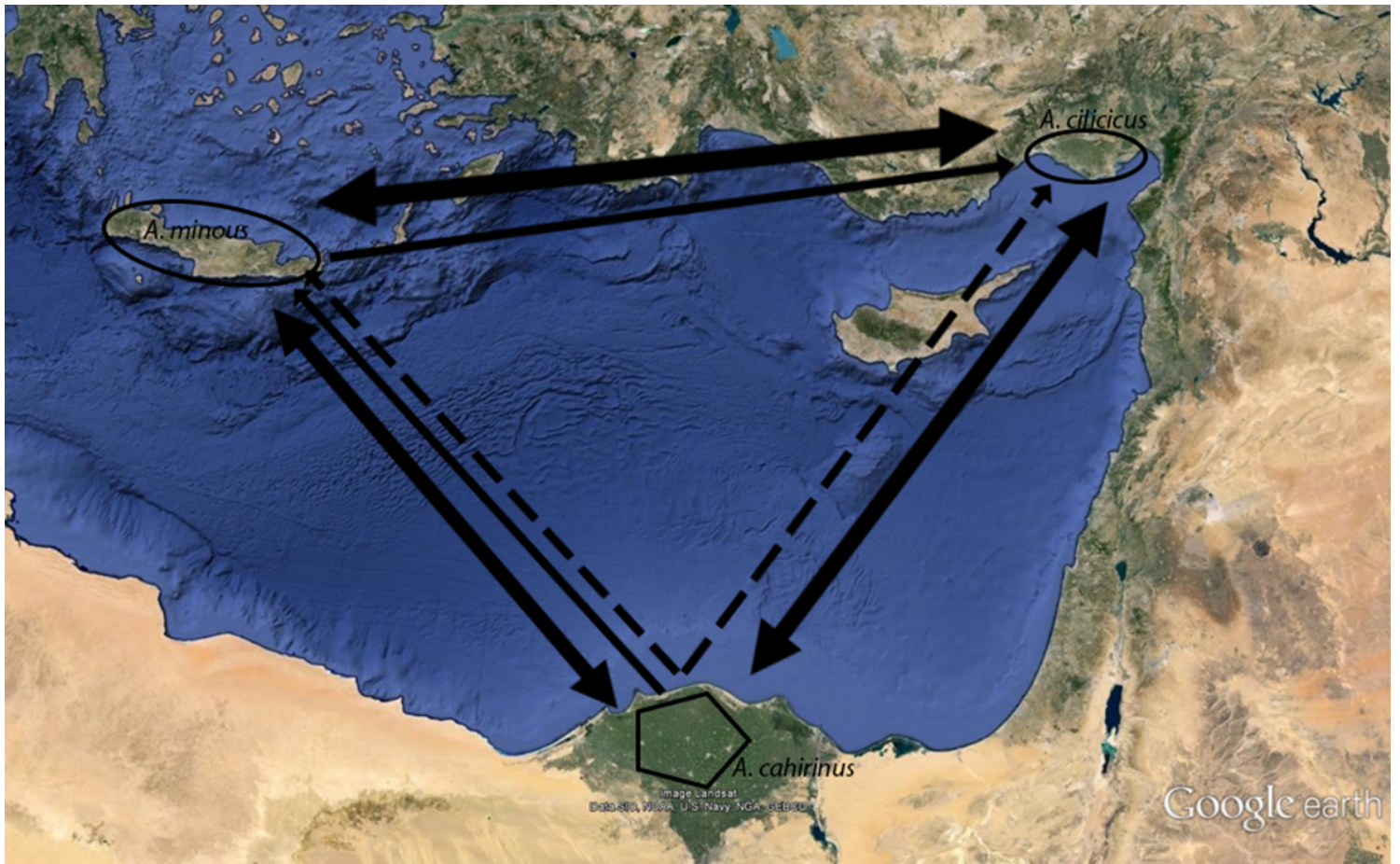


Figure 4

Schematic representation of 3 possible migration routes tested by MIGRATE. Model 1 is shown with dashed arrows, model 2 with single headed arrows, and model 3 with double headed arrows. According to model 3, migration paths are bidirectional and the ancestral status of *Acomys cahirinus* is doubtful. Models 1 and 2 were constructed with the assumption that *A. cahirinus* was an ancestral species. According to these models, migration routes are unidirectional.

Supplementary Files

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